Operation Komorebi

Large Signal Model

Visual notes on certified probability domains

Seven mathematical views.
Exact witnesses and explicit error budgets.
All figures regenerated from Python source.

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Foundations edition 0.4.0

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What these figures establish

These notes accompany the fourth foundations block, Propositions 38–49. The figures describe probability-domain certificates and their approximation cost. They are mathematical views of declared models, not task-suite results.

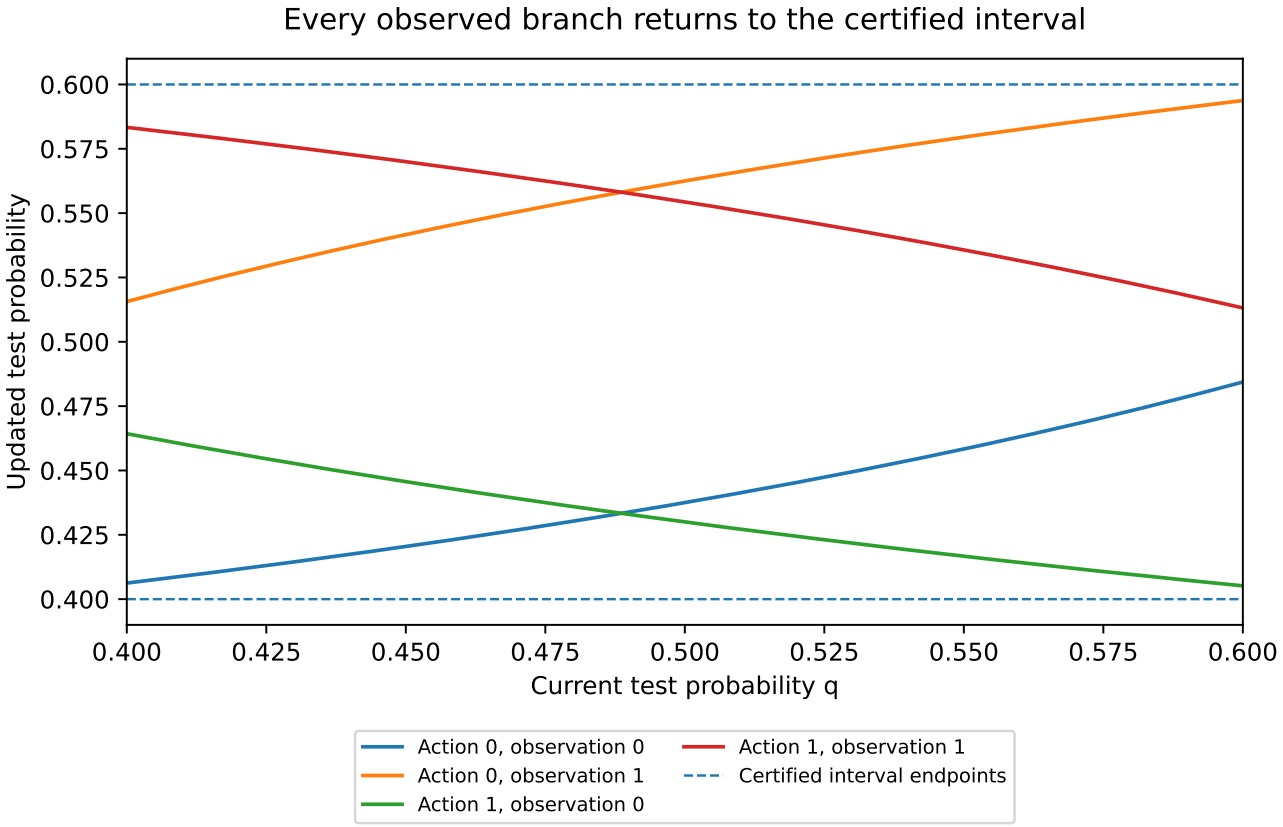
A learned operator family is a stochastic model only on a domain where its outputs remain probability-valid. The finite witness in the first two views is an exact rational identity. Curves illustrate its action. They are not the reason the identity holds.

The full-law figures distinguish a proved horizon-dependent bound from the exact finite law calculated for one example. The curved-domain figure illustrates a general obstruction whose proof uses an irrational rotation. No finite scatter plot proves density of an orbit.

All seven new figures have Python source and separate PDF, PNG and SVG assets. Exact plotted quantities are recorded as rational values before conversion to display coordinates. The regular-polygon curve is a numerical display of a stated analytic formula. The cumulative volume supplies the complete hypotheses and proofs.

Read the distinction throughout: a valid fitted model, an accurate model of observed data, and a statistically justified model of the generating process are three different conclusions.

1. A domain found from the observable operators

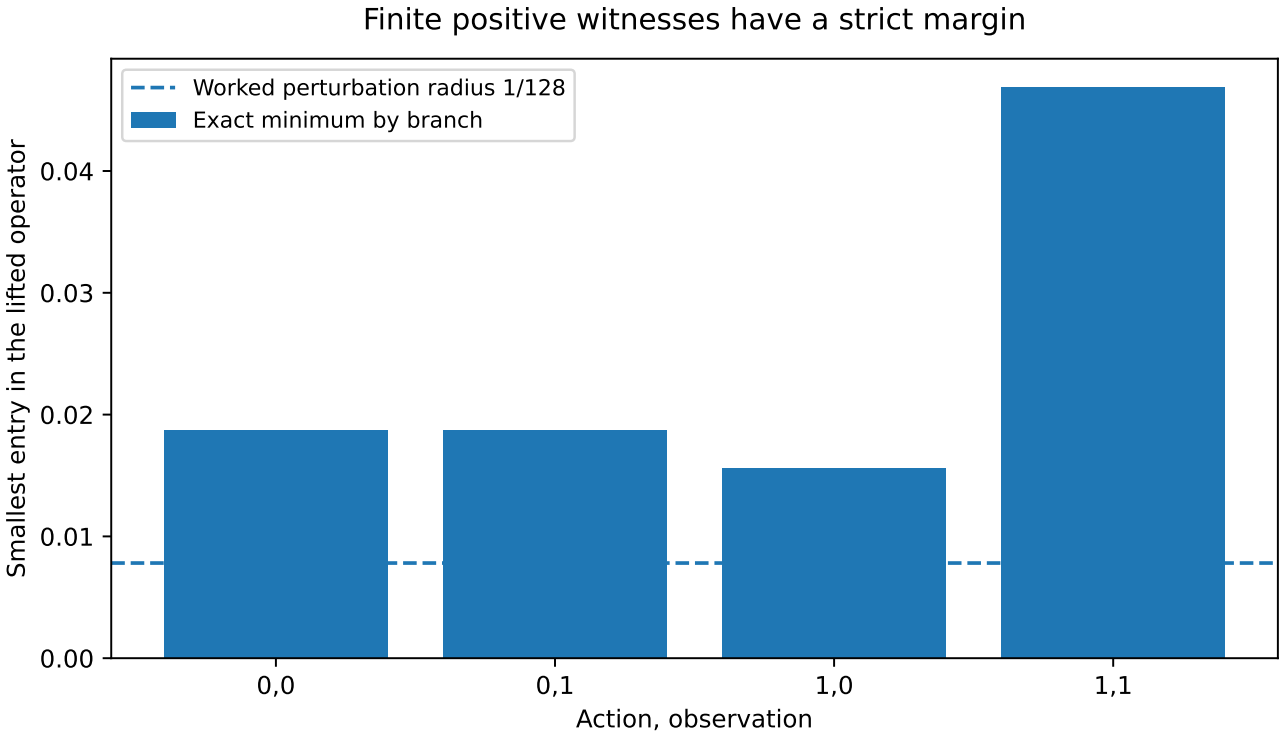


The rational interval search finds $2/5 \leq q \leq 3/5$. Every one of the four action and observation branches maps this interval back into itself after positive normalization. The initial coordinate is $q_0 = 13/24$.

The search receives the operators obtained from the observable probability table, not hidden-state matrices. Exact nonnegative lifts certify invariance for every point, not just the plotted grid. The displayed interval is the first accepted candidate in a specified finite rational grid. It is not claimed to be the smallest invariant interval among all real endpoints.

Contract: a finite checked witness supports all-horizon internal probability validity. Global rank coverage and accuracy of the population table remain separate assumptions.

2. A positive lift gives a robust margin

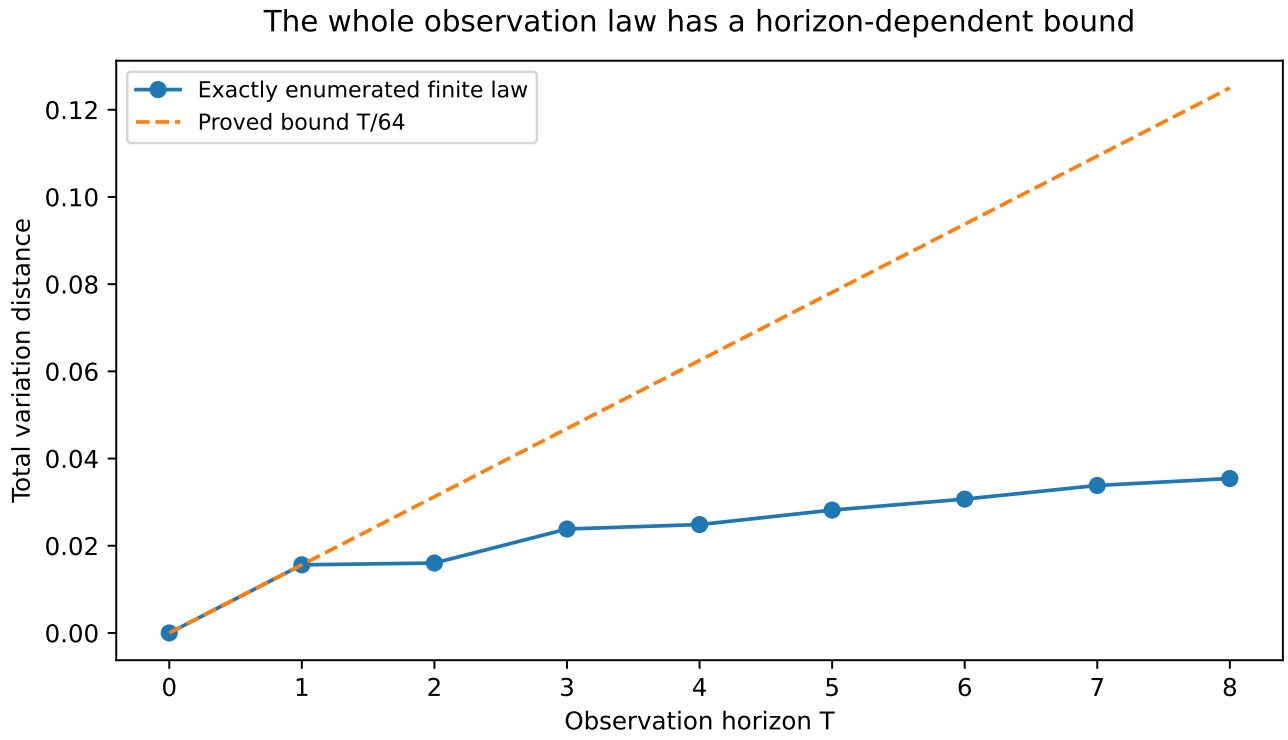


With $V = \begin{pmatrix} 1 & 1 \\ 2/5 & 3/5 \end{pmatrix}$, every lifted operator $A_s = V^{-1} B_s V$ is nonnegative. The smallest entry across all branches is exactly $1/64$. The initial positive weights are $(7/24, 17/24)^\top$.

Perturbing each lifted entry by at most $1/64$ preserves nonnegativity, provided the perturbations are coupled to preserve the per-action column sums. The worked perturbations use the smaller radius $1/128$.

Contract: the margin concerns this coordinate lift and mass-conserving uncertainty family. An unrestricted box around all predictive-matrix entries is not automatically a family of normalized probability models.

3. Bound the complete observation law before conditioning



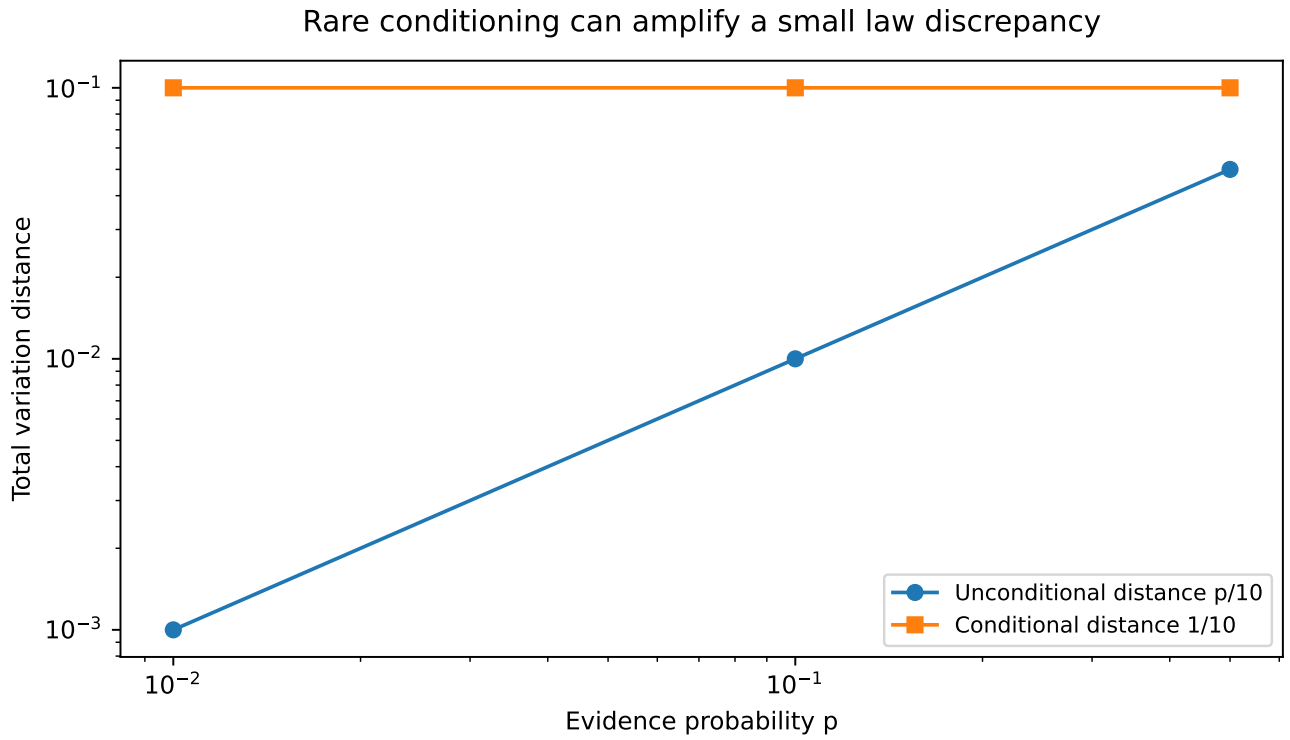
The two valid models share the certified cone and the same history-dependent action policy. Their initial states coincide. The summed base-norm discrepancy is at most $\epsilon = 1/32$, giving

$$\|P_T - \widetilde{P}_T\|_{\mathrm{TV}} \leq \min(1, T/64).$$

The lower curve is the exact finite law obtained by summing rational word probabilities through horizon eight. The upper curve is the general bound applied to the declared example. No rare-branch likelihood floor is required for this whole-law comparison.

Contract: the bound grows with horizon. It is not a uniform-in-time theorem or a comparison of separately optimized policies.

4. Rare conditioning has a different sensitivity



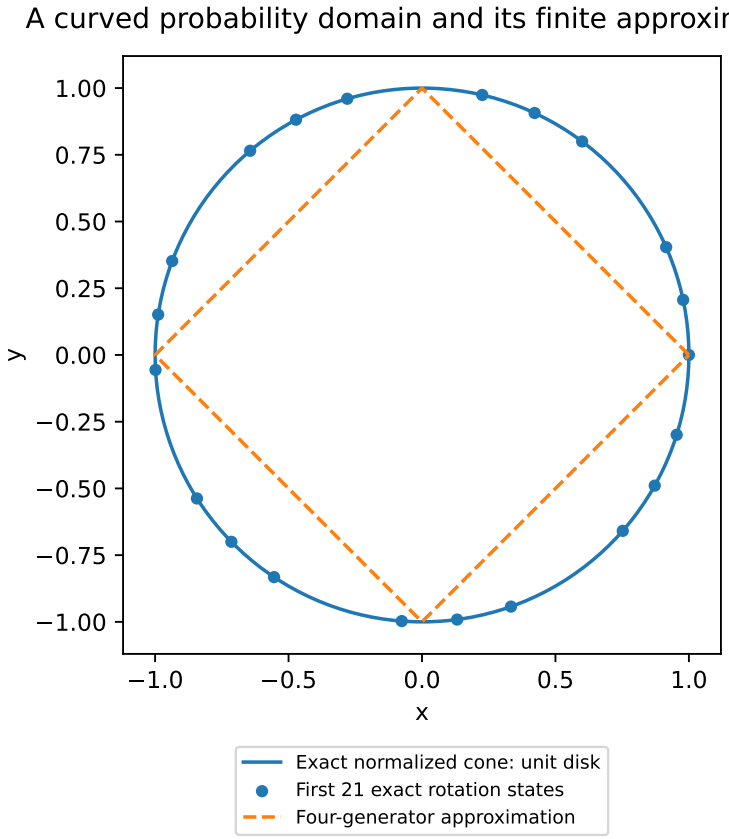
Each example moves probability mass $\delta = p/10$ between two outcomes inside an event of probability p. The unconditional total variation is δ, while the conditional distance is exactly $1/10$.

$$\|P(\cdot \mid H) - Q(\cdot \mid H)\|_{\mathrm{TV}} \leq \min(1, \delta/P(H)).$$

As the event becomes rarer, a smaller unconditional discrepancy can create the same conditional change. This is a property of conditioning, not evidence that the underlying probability model is invalid.

Contract: a whole-law error bound and a posterior stability bound answer different questions. The event probability must remain in the latter certificate.

5. Three predictive coordinates can need a curved domain

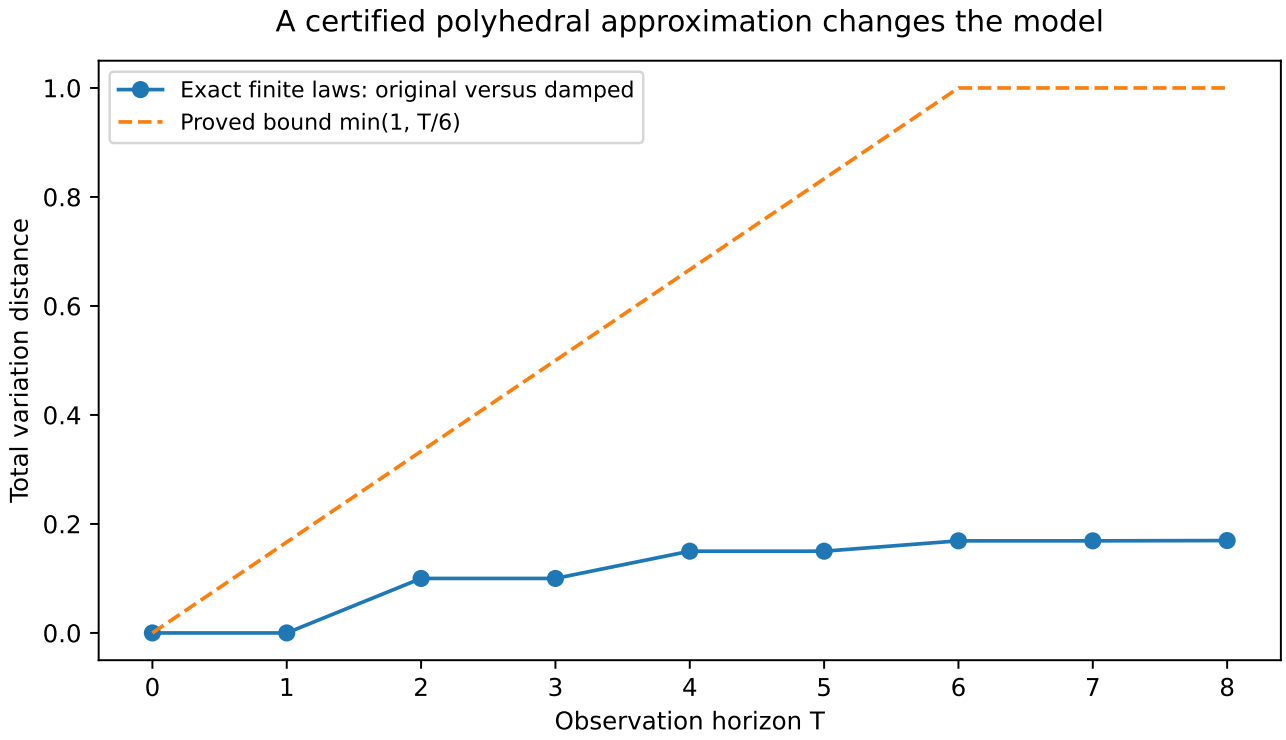


The normalized exact probability domain is the unit disk in the (x, y) plane. The mass coordinate is fixed at $t = 1$. One action rotates by the rational matrix with cosine $3/5$ and sine $4/5$. Probe actions expose all rotated directions.

The proof shows that the rotation orbit is dense and that every closed convex invariant probability domain containing the initial state must be this disk. A finite polygon therefore cannot certify the exact model. The plotted points are only a finite illustration of that proof.

Contract: this is a boundary for polyhedral certificates, not for all finite certificates. The disk has the finite quadratic description $x^2 + y^2 \leq 1$.

6. An explicit approximation restores a finite positive lift

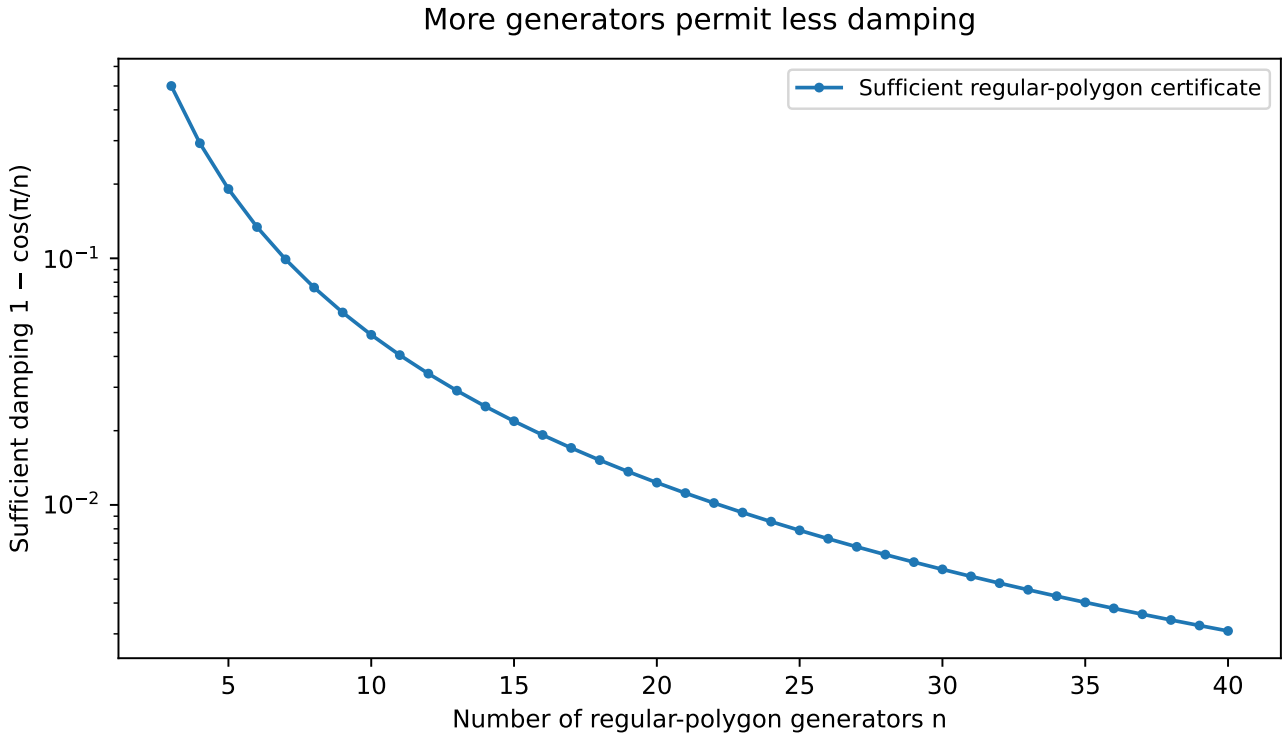


Replacing the planar rotation R by $(2/3)R$ makes the four-vertex diamond invariant. All four generators and their positive-lift witnesses are rational. The undamped model does not preserve that diamond.

Both models remain valid on the larger exact disk. Their summed instrument discrepancy is at most $1/3$, giving the whole-law bound $\min(1, T/6)$ under the same policy. The finite law curve uses alternating rotate and probe actions.

Contract: damping changes the model. The certificate records the modification and bounds its probability-law cost. It is not described as lossless refinement.

7. Finite representation size has an approximation cost



A regular n-gon inscribed in the disk contains the disk of radius $\cos(\pi/n)$. Consequently a rotation damped by $\alpha \leq \cos(\pi/n)$ preserves its cone, giving a finite positive realization.

For a target damping $\eta = 1 - \alpha$, the sufficient choice

$$n \geq \max\left(3, \left\lceil \frac{\pi}{\sqrt{2\eta}} \right\rceil\right)$$

follows from $1 - \cos u \leq u^2/2$. This is an upper bound from one construction, not an optimal generator count. The polygon may have irrational vertices. The separate diamond example supplies an exact rational instance.

Contract: the geometric representation, the change in dynamics, and the finite-horizon prediction bound must be accounted for together.